

Part I

Recitation Questions

Problem 1

February 21, 2026

Problem 1

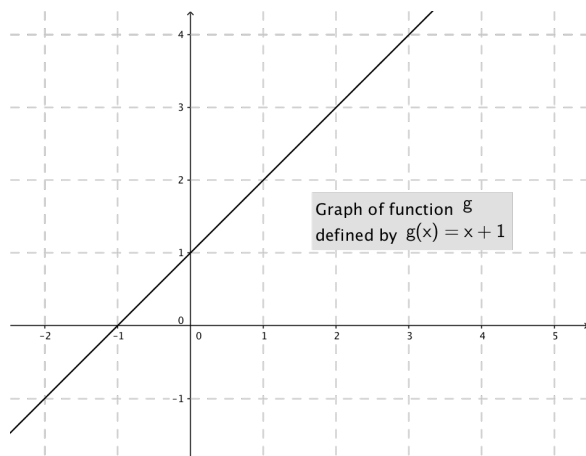
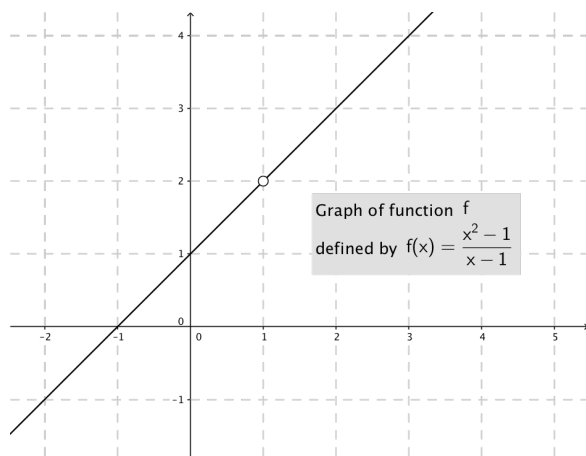
- (a) *True or False: To find $\lim_{x \rightarrow 2} f(x)$, it's enough to know the values of $f(2.1)$, $f(2.01)$, $f(2.001)$, and so on.*
- (b) *True or False: If we know the value of $f(2)$, then can we conclude that $\lim_{x \rightarrow 2} f(x) = f(2)$?*

Problem 2

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Problem 2

Problem 2 Use the graphs and the given definitions of the following two functions to answer the questions below.



- Find the domain of f and the domain of g .
- Is $f = g$? (Why or why not?)
- Looking at the graphs, find $\lim_{x \rightarrow 1} f(x)$ and $\lim_{x \rightarrow 1} g(x)$.

Problem 3

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Problem 3

Problem 3 The graph of a function f is given below. Use this graph to answer the following questions.

alt=graph of piecewise-defined function for finding domain, range, and limit values from graph.

(a) Find the domain of f .

(b) Find the range of f .

(c) Find the following values:

(i) $\lim_{x \rightarrow -2} f(x) =$

(ii) $f(-2) =$

(iii) $f(-5) =$

(iv) $\lim_{x \rightarrow 0^+} f(x) =$

(v) $\lim_{x \rightarrow 0} f(x) =$

Problem 4

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Problem 4

Problem 4 Sketch a possible graph of a function that satisfies all of the given properties. (You do not need to find a formula for the function.)

$$\begin{array}{llll} \text{Domain: } [-4, 3) \cup (3, 5) & f(1) = 3 & f(-1) = 1 & f(-4) = 1 \\ \lim_{x \rightarrow -1^-} f(x) = -2 & \lim_{x \rightarrow -1^+} f(x) = 1 & \lim_{x \rightarrow 1} f(x) = 2 & \lim_{x \rightarrow 3^+} f(x) = -1 \\ \lim_{x \rightarrow 3^-} f(x) = 1 & & \lim_{x \rightarrow -4^+} f(x) = 1 & \lim_{x \rightarrow 5^-} f(x) = 1 \end{array}$$

Problem 5

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Problem 5

Problem 5 True/False: Give an explanation or counterexample. Assume a and L are finite numbers.

- (a) If $\lim_{x \rightarrow a} f(x) = L$, then $f(a) = L$.
- (b) If $\lim_{x \rightarrow a^-} f(x) = L$, then $\lim_{x \rightarrow a^+} f(x) = L$.
- (c) If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = L$, then $f(a) = g(a)$.
- (d) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ does not exist if $g(a) = 0$.